

AI LAB 3.py - C:/Users/K Aishwarya/Documents/AI LAB 3.py (3.14.2)

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from collections import deque

def water_jug():
    a, b = 0, 0
    visited = set()
    queue = deque([(a, b), []])

    while queue:
        (x, y), path = queue.popleft()

        if (x, y) in visited:
            continue

        visited.add((x, y))
        path = path + [(x, y)]

        if x == 2:
            print("Solution:")
            for state in path:
                print(state)
            return

        states = [
            (4, y),
            (x, 3),
            (0, y),
            (x, 0),
            (x - min(x, 3 - y), y + min(x, 3 - y)),
            (x + min(y, 4 - x), y - min(y, 4 - x))
        ]

        for state in states:
            if state not in visited:
                queue.append((state, path))

water_jug()
```

```
>>>|
===== RESTART: C:/Users/K Aishwarya/Documents/AI LAB 3.py =====
Solution:
(0, 0)
(4, 0)
(1, 3)
(1, 0)
(0, 1)
(4, 1)
(2, 3)
>>>|
```